

REVIEW 4 · WORKING PROTOTYPE

PlagiaScan

Dashboard

● System Ready

PlagiaScan

Dashboard

Documents

Comparisons

Reports

An Explainable Multi-Algorithm Plagiarism & Document Similarity Analyzer

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DOCUMENT INTELLIGENCE

Analyze document similarity with clarity.

PlagiaScan combines multiple similarity techniques with explainable evidence to show exactly where documents overlap.

Add Documents →

D

DOCUMENTS

0

C

COMPARISONS

0

%

HIGHEST
SIMILARITY

—

✓

STATUS

Ready

Made with GAMMA

● System Ready

THE PROBLEM

Similarity is harder than copying

Forms of copying

Plagiarism can involve exact copying, partial reuse, word changes, rearrangement, or paraphrasing.

A score is not an explanation

A single percentage does not explain why two documents are similar.

Documents add scale

Manual comparison across multiple documents is slow and difficult to scale.

Make evidence visible

PlagiaScan makes the evidence visible alongside the score.

Documents → forms of copying → difficult comparison → explainable automation

THE PRODUCT PROMISE

PlagiaScan turns comparison into evidence

01

Input documents

Compare multiple documents simultaneously.

02

Multi-algorithm analysis

Measure lexical, exact, structural, and semantic similarity.

03

Evidence extraction

Extract matching phrases and algorithm-level evidence.

04

Similarity fusion

Fuse complementary signals into one overall plagiarism similarity score.

05

Explainable report

Return rankings, matrices, reports, and comparison history.

IMPLEMENTED ARCHITECTURE

The system separates ingestion, analysis, and explanation

React UI ↔ REST API ↔ Java Analysis Engine

FRONTEND

React + Vite

Dashboard, document upload, comparisons, reports and comparison history.

Persistence: LocalStorage

API + BACKEND

Javalin REST API

Java 21 + Maven analysis service for upload, pairwise comparison and report delivery.

Modular lexical, exact, edit-distance and semantic signals.

SEMANTIC ENGINE

DJL + Hugging Face

Model: all-MiniLM-L6-v2 for meaning-level similarity and explainable score fusion.

Persistence and history remain in the frontend.

No single algorithm covers every form of plagiarism

Complementary algorithms are stronger than one metric alone.

Word Overlap

Common vocabulary and lexical similarity.

KMP

Repeated exact phrases through pattern matching.

Rabin-Karp

Matching token sequences through hashing.

Token Edit Distance

Insertions, deletions, and substitutions.

MiniLM + Cosine Similarity

Meaning-level and paraphrased similarity.

From uploaded documents to pairwise evidence

01

Ingest

Upload multiple documents. Extract and normalize text: lowercase, clean characters, normalize whitespace, and tokenize.

02

Compare

Run lexical and exact phrase comparison; calculate token-level edit distance.

03

Explain

Generate semantic embeddings with MiniLM; calculate cosine similarity; fuse scores into an explainable report.

Lexical and exact branches converge with edit-distance and semantic branches before pairwise reporting.

SCORING MODEL

The final score is a weighted combination of signals

TEXTUAL SIMILARITY

Word Overlap = $(|A \cap B| / \max(|A|, |B|)) \times 100$

Edit Similarity = $\max(0, (1 - d/L) \times 100)$, where d is token edit distance and L is maximum token length.

Textual Similarity = $0.40 \times \text{Word Overlap} + 0.30 \times \text{Edit Similarity} + 0.30 \times \text{Exact Match Coverage}$.

SEMANTIC SIMILARITY

Cosine similarity over all-MiniLM-L6-v2 embeddings.

FINAL SIMILARITY

Final Similarity = $0.70 \times \text{Textual Similarity} + 0.30 \times \text{Semantic Similarity}$.

EXPLAINABLE OUTPUT

Beyond a Single Percentage

PlagiaScan explains **WHY** two documents are similar — not just **HOW** similar they are.

01 • MULTI-METRIC

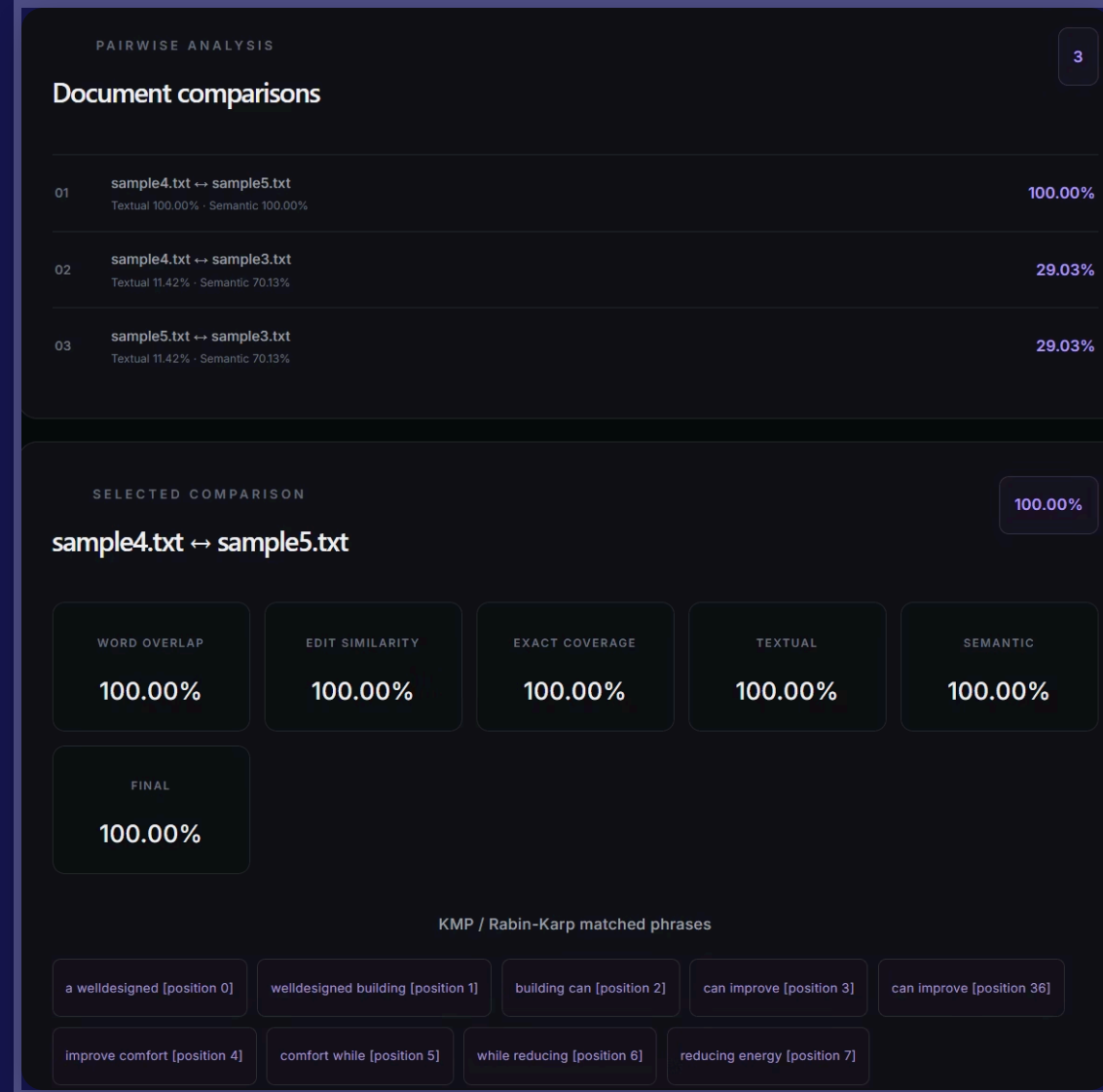
Word overlap, edit similarity, exact coverage, textual similarity and semantic similarity.

02 • ALGORITHM EVIDENCE

KMP and Rabin-Karp identify matching phrases and their positions.

03 • EXPLAINABLE RESULT

The system shows the evidence behind the final plagiarism similarity score.



Actual PlagiaScan working prototype — pairwise analysis with explainable evidence

PROTOTYPE RESULTS

Working Results

Pairwise analysis from the implemented PlagiaScan prototype

SIMILARITY SCORES

sample4.txt ↔ sample5.txt · Textual 100.00% · Semantic 100.00% · Overall 100.00%

sample4.txt ↔ sample3.txt · Textual 11.42% · Semantic 70.13% · Overall 29.03%

sample5.txt ↔ sample3.txt · Textual 11.42% · Semantic 70.13% · Overall 29.03%

HIGHEST-SIMILARITY PAIR

sample4.txt ↔ sample5.txt

Overall Similarity: 100.00%

EVIDENCE

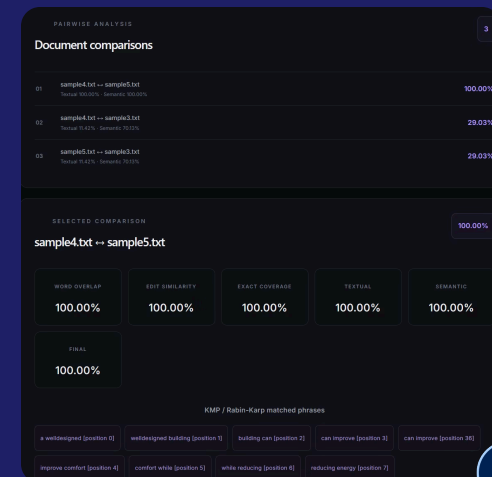
Word Overlap

Edit Similarity

Exact Coverage

KMP matched phrases

Rabin-Karp matched phrases



ACCURACY OF RESULTS

Accuracy of Results

Multiple independent signals make the result easier to verify — without inventing statistical accuracy claims.

Input Documents → Word Overlap → Edit Distance → KMP → Rabin-Karp → Semantic Similarity → Weighted Score → Explainable Evidence

WORKING PROTOTYPE SANITY CHECK – IDENTICAL DOCUMENTS

Word Overlap 100% · Edit Similarity 100% · Exact Coverage 100% · Textual Similarity 100% · Semantic Similarity 100% · Final Similarity 100%

This identical-document check is a prototype sanity check; it does not mean 100% detection accuracy.

What Makes PlagiaScan Different?

01 Multi-algorithm detection

Combines lexical, exact, sequence and semantic similarity.

02 Explainable results

Shows matched phrases and algorithm-level evidence.

03 Semantic plagiarism detection

MiniLM identifies meaning-level similarity beyond exact wording.

04 Multi-document analysis

Pairwise comparisons, rankings and similarity matrices.

05 Lightweight software solution

No specialized hardware is required.



Sustainability: Digital plagiarism screening reduces repetitive manual comparison and supports paperless document evaluation.

UTILITY TO SOCIETY

Utility to Society

Students

Assignment checking

Faculty

Submission screening

Universities

Academic integrity

Researchers

Document comparison

Content Creators

Duplicate content detection

Educational Platforms

Automated similarity screening

PlagiaScan helps make academic document evaluation faster, more transparent and evidence-based.

Review 4 — Working Prototype

From similarity score → to explainable evidence.

- ✓ Multi-document comparison
- ✓ Multi-algorithm analysis
- ✓ Semantic similarity
- ✓ Explainable matching evidence
- ✓ Similarity matrix
- ✓ Pairwise ranking
- ✓ Reports
- ✓ Comparison history

Thank You